

In this assignment, we had to print TCP SACK (selective Acknowledgements) which are ranges of acknowledgements received successfully at the receiver. This is the method used instead of just sending the next expected seqno by the sender used in TCP.

For this we firstly enabled TCP sack on the receiver. Then wrote code in pcap.h file which reads the optional header. In it, we skip if SACK kind=1, terminate if kind=0, and skip len bytes if kind!=5, otherwise if kind=5, we read the len bytes. The number of SACK acknowledgements is $(len-2)/8$. I manually loop over this number reading by `ntohl(l)`, `ntohl(r)` of ranges as unsigned integers. Print the range with the count number of this TCP sack in the entire duration.

I am running the experiments for a 110 Mb file instead of 1Gb because that takes approximately 40 minutes to transfer through SCP.

- 1) 3mintues, 37 seconds.
- 2) 16minutes, 34 seconds approximately.
- 3) 7845 SACKs received in total
- 4) 7335 range 1 SACKS, 250 range 2 SACKS, 201 range 3 SACKS
- 5) I encountered kinds 3,4,8,5 with 8 being the most frequent.

- a.) kind 3 main purpose is to increase the maximum window size from 65,335 bytes to upto 1gb.
- b.) kind 4 main purpose is to inform the other host via signal that the device is capable of handling selective acknowledgments.
- c.) kind 5 is basically used to notify the sender the ranges of data received by the receiver .
- d.) kind 8 main purpose to allow for more accurate (RTT) measurements and to prevent data corruption when very large window cause sequence numbers to wrap around.

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Kind-1
Kind-8
Kind-1
Kind-1
Kind-5
Range 4234 192854806-192857702, 1->3254, 2->214, 3
Kind-1
Kind-1
Kind-8
Kind-1
Kind-1
Kind-8
Kind-1
Kind-1
Kind-5
Range 4235 192854806-192859150, 1->3255, 2->214, 3
Kind-1
Kind-1
Kind-8